Introduction to Kriging (Handbook of Spatial Statistics, Chapter 3)

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Motivation

Prediction in the form of interpolation is of high importance to many geological disciplines, where oftentimes only a small number of spatially-indexed samples can be taken, yet information is needed about other areas.

Most general interpolation methods rely only on distance to "weight" interpolations, and effectively lose information from spatial autocorrelation.

Kriging seeks to change this by using both distance and autocorrelation to interpolate. Additionally, kriging creates a means of assessing the accuracy of our interpolations.

Types of Kriging

- *Simple* Kriging (assumes constant mean = average of output values)
- *Ordinary* Kriging (assumes a constant mean, estimated by generalized least squares estimate of μ)
- *Universal* Kriging (does not assume a constant mean, modeled by linear regression with spatially-referenced covariates)

We will focus on Universal Kriging.

History

Ordinary kriging was developed in 1951 by Danie G. Krige (1919-2013), a South African statistician, mining engineer, and later Professor at University of the Witwatersrand. Krige originally used his technique for mineral exploration.

Krige's ideas were formalized in 1962 by Georges Matheron.

Setup

Suppose we measure Y at a series of points $s = (s_1, \ldots, s_n)$, where $s_i \subset D$ $i = 1, \ldots, n$, $D = \mathbb{R}^2$ or $\mathbb{R}^3$. And, we write

$$Y(s) = \mu(s) + e(s)$$

where $\mu(s) = E[Y(s)]$ and $e(\cdot) = \{e(s)|s \in D\}$ is a zero-mean random error process satisfying a stationarity assumption. Furthermore, we define

$$e(s) = \eta(s) + \epsilon(s)$$

where $\eta(s)$ is the spatially-dependent component and $\epsilon(s)$ is a random measurement error.

Specify Parametric Model for μ

We begin by specifying a parametric model for μ. This nearly always takes the form of a linear regression model,

$$\mu(s|\beta) = X(s)^T \beta$$

where $X(s)$ is a vector of covariates and β an unrestricted parameter vector.

The standard method for fitting the mean function is OLS:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y$$

so $\hat{Y} = X^T \hat{\beta}_{OLS}$ and $\hat{e} = Y - \hat{Y}$ (important for next step)

Notes on Covariates

Because our intention is to interpolate, we will ultimately need data from each covariate at every point in our space D that is of interest.

Thus, covariates in the mean model are typically

- coordinates (latitude and longitude),
- mathematical functions of those coordinates (e.g., polynomials),
- easily-referenced physical attributes (e.g., elevation, average wind speed, average temperature, etc.).

If only the coordinates and/or functions of the coordinates are used as covariates, the mean model is called a *trend surface model*.

Nonparametric Estimation of the Semivariogram

Assume we are working in $D = \mathbb{R}^2$, and consider two scenarios:

(1) If our spatial data forms a regular rectangular grid, let

$$h_1 = (h_{11}, h_{12})^T, \ldots, h_k = (h_{k1}, h_{k2})^T$$

represent the k distinct lags between points $s \in D$, with displacement angles

$$\phi(u) = \tan^{-1}(h_{u2}/h_{u1}) \in [0, \pi)$$

Then the *empirical semivariogram* is defined as

$$\hat{\gamma}(h_u) = \frac{1}{2N(h_u)} \sum_{s_i - s_j = h_u} \{\hat{e}(s_i) - \hat{e}(s_j)\}^2, \ u = 1, \ldots, k$$

where $N(h_u)$ is the number of times the lag h_u occurs among the data locations.

Nonparametric Estimation of the Semivariogram (cont.)

(2) When our data is irregularly spaced, we partition the set of sample spacings into k bins, H_1, H_2, ..., H_k, and define the empirical semivariogram as

$$\hat{\gamma}(h_u) = \frac{1}{2N(H_u)} \sum_{s_i - s_j \in H_u} \{\hat{e}(s_i) - \hat{e}(s_j)\}^2, \ u = 1, \ldots, k$$

How many bins? "Rule of Thumb" given as $N(H_u) \geq 30$ and length of $H_u <$ half the maximum lag length $\forall u = 1, \ldots, k$

Model the Semivariogram

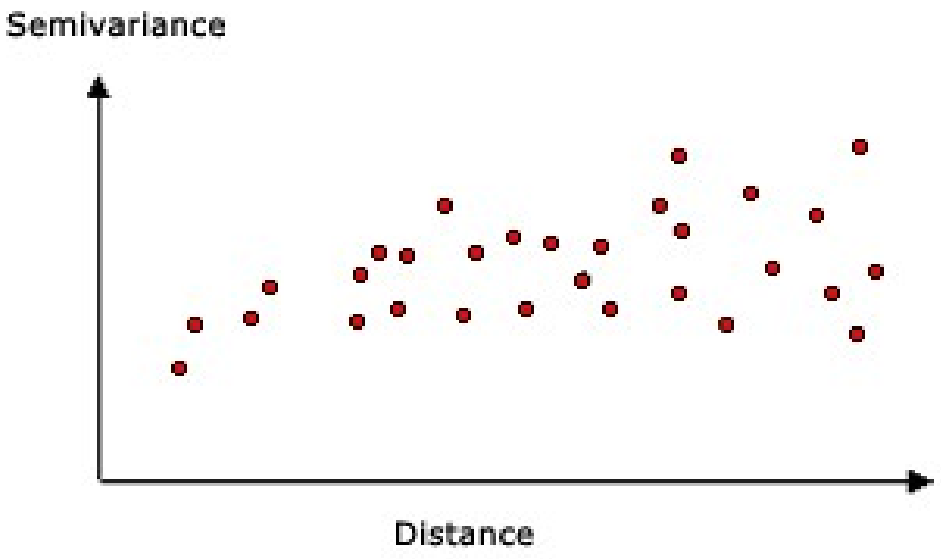


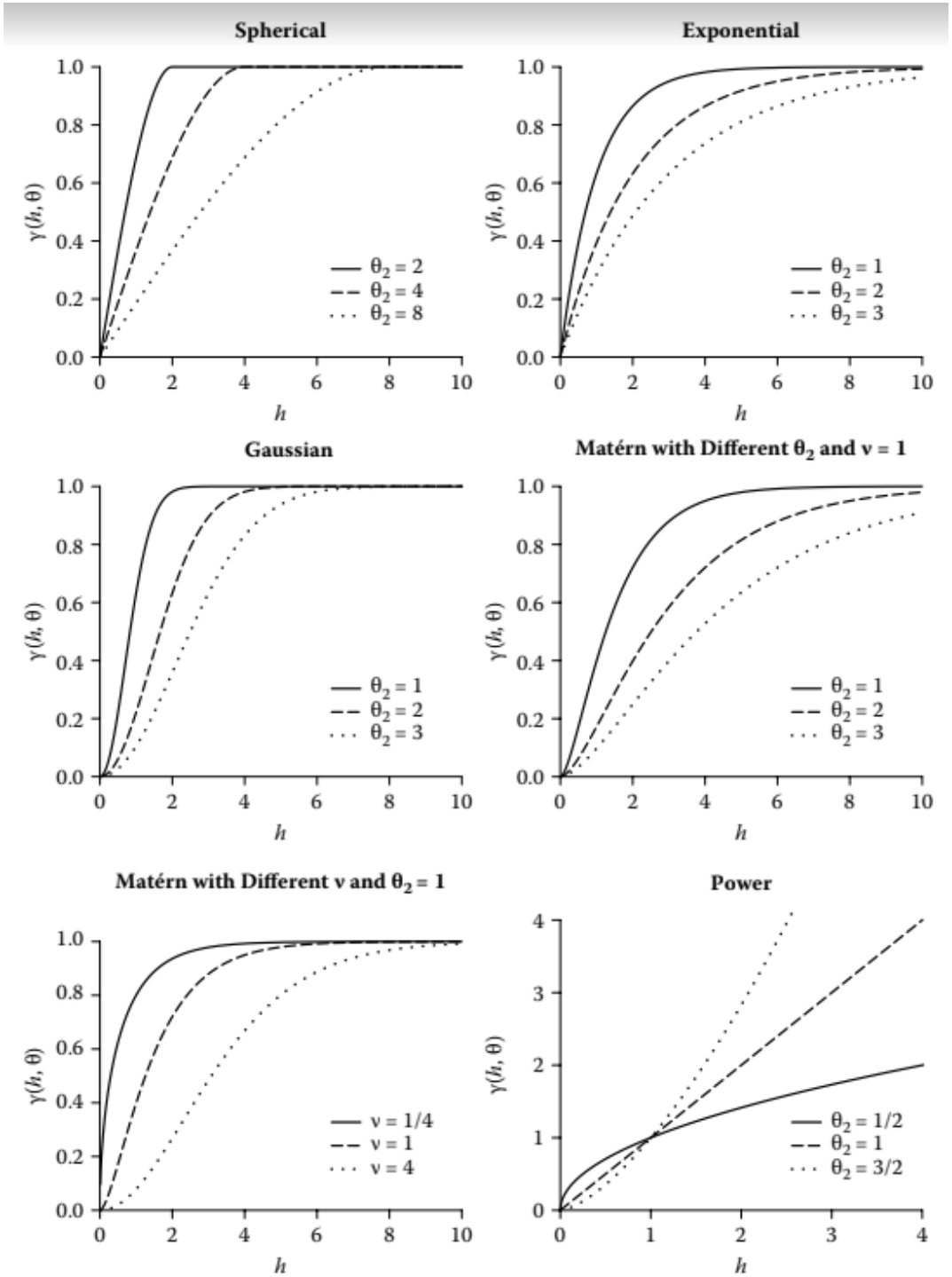
Figure 1: Example Semiovariogram

Once we have calculated the semivariance for each unique sample spacing (or bin of sample spacings), we fit a model through their plot.

Model the Semivariogram (cont.)

The following three conditions are necessary and sufficient for a semivariogram model, $\gamma(h|\theta)$ to be valid, provided that they hold for all θ in the parameter space for θ:

1. Vanishing at 0, i.e., $\gamma(0|\theta) = 0$
2. Evenness, i.e., $\gamma(-h|\theta) = \gamma(h|\theta)\ \forall h$
3. Conditional negative definiteness, i.e., $\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \gamma(s_i - s_j|\theta) \leq 0 \quad \forall n, \forall s_i, \forall a_i$ such that $\sum_{i=1}^{n} a_i = 1$


Spherical
Exponential
Gaussian
Matérn with Different θ2 and ν = 1
Matérn with Different ν and θ2 = 1
Power
γ(h, θ)
h
θ2 = 2
θ2 = 4
θ2 = 8
θ2 = 1
θ2 = 2
θ2 = 3
ν = 1/4
ν = 1
ν = 4
θ2 = 1/2
θ2 = 1
θ2 = 3/2

Model the Semivariogram (cont.)

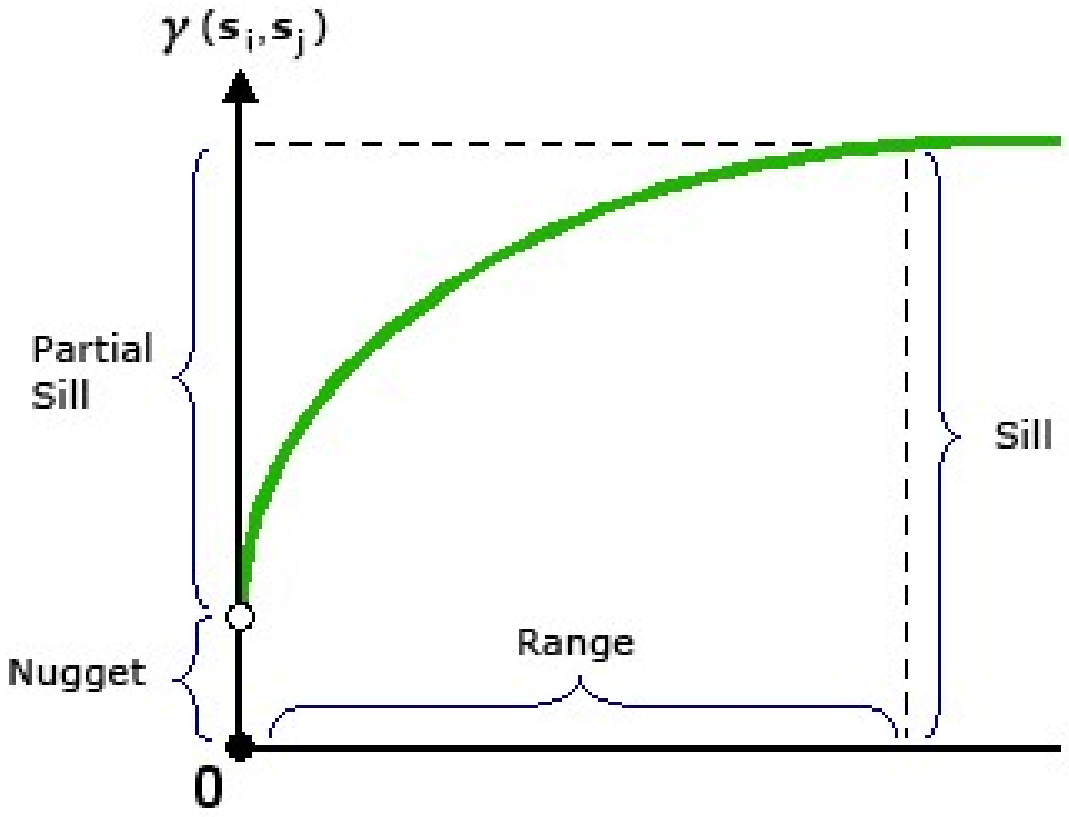
The parameter(s) θ is usually fit using weighted least squares (WLS):

$$\hat{\theta}_{WLS} = argmin \sum_{u} \frac{N(h_u)}{[\hat{\gamma}(h_u|\theta)]^2} [\hat{\gamma}(h_u) - \gamma(h_u|\theta)]^2$$

WLS is most common due to its relative simplicity, but is noted to be suboptimal because it is not based on an underlying probabilistic model for the spatial process. Common alternatives are maximum likelihood or residual maximum likelihood.

However, WLS yields consistent and asymptotically normal estimators under certain regularity conditions and is generally not inferior to likelihood-based methods (Lahiri, Lee and Cressie (2002); Zimmerman and Zimmerman (1991); Lark (2000)).

Model the Semivariogram (cont.)



$Sill = \lim_{h\to\infty} \gamma(h|\theta)$
$Nugget = \lim_{h\to 0} \gamma(h|\theta)$
Range: Smallest value of h for which $\gamma(h|\theta) = \text{Sill}$

Kriging

Universal kriging is the best linear unbiased predictor (BLUP).

Let $s_0 \in D$, where s_0 is usually an unsampled point but need not be. We want to find a predictor, $\hat{Y}(s_0)$ with the minimal prediction error variance, $var[\hat{Y}(s_0) - Y(s_0)]$, among all predictors satisfying:

1. Linearity $\hat{Y}(s_0) = \lambda^T Y$, where λ is a vector of fixed constants
2. Unbiasedness $E[\hat{Y}(s_0)] = E[Y(s_0)]$

Kriging (cont.)

"The solution to this constrained optimization problem, known as the universal kriging predictor of $Y(s_0)$, is given by

$$\hat{Y}(s_0) = [\gamma + X(X^T\Gamma^{-1}X)^{-1}(x_0 - X^T\Gamma^{-1}\gamma)]^T\Gamma^{-1}Y$$

where $\gamma = [\gamma(s_1 - s_0), \ldots, \gamma(s_n - s_0)]^T$, Γ is the $n \times n$ symmetric matrix with ij^{th} element $\gamma(s_i - s_j)$ and $x_0 = X(s_0)$."

Common Modifications:

- To reduce computation, replace the entire vector Y with a subset of points in a neighborhood around $Y(s_0)$
- Substitute γ and Γ with $\hat{\gamma} = \gamma(\hat{\theta})$ and $\hat{\Gamma} = \Gamma(\hat{\theta})$, respectively

Kriging (cont.)

The *universal kriging variance* is the minimized value of the prediction error variance, given by

$$\sigma^2(s_0) = \gamma^T \Gamma^{-1} \gamma - (X^T \Gamma^{-1} \gamma - x_0)^T (X^T \Gamma^{-1} X)^{-1} (X^T \Gamma^{-1} \gamma - x_0)$$

If $Y(\cdot)$ is Gaussian, then the above variance can be used to created a $(1-\alpha)$ prediction interval for $Y(s_0)$, given by

$$\hat{Y}(s_0) \pm z_{\alpha/2} \times \sigma(s_0)$$

where $0 < \alpha < 1$ and $z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of a standard normal distribution.

Discussion

1. How can we interpret the nugget? Sill? Range?
2. What is the impact of model specification for the semivariogram on the final predicted values?
3. Similarly, what is the impact of model specification for μ? Does including more covariates likely increase prediction accuracy?

References

- A. Gelfand, M. Fuentes, P. Guttorp, and P. Diggle, Handbook of Spatial Statistics. Chapman Hall/CRC Handbooks of Modern Statistical Methods, Taylor & Francis, 2010 (29-44).
- "How Kriging Works". ESRI. 2016. http://desktop.arcgis.com/en/arcmap/10.3/tools/3d-analyst-toolbox/how-kriging-works.htm